SAP CPI Project Documentation

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Scenario: Idempotent Process Call IFlow in SAP Integration Suite

1.Scenario Overview

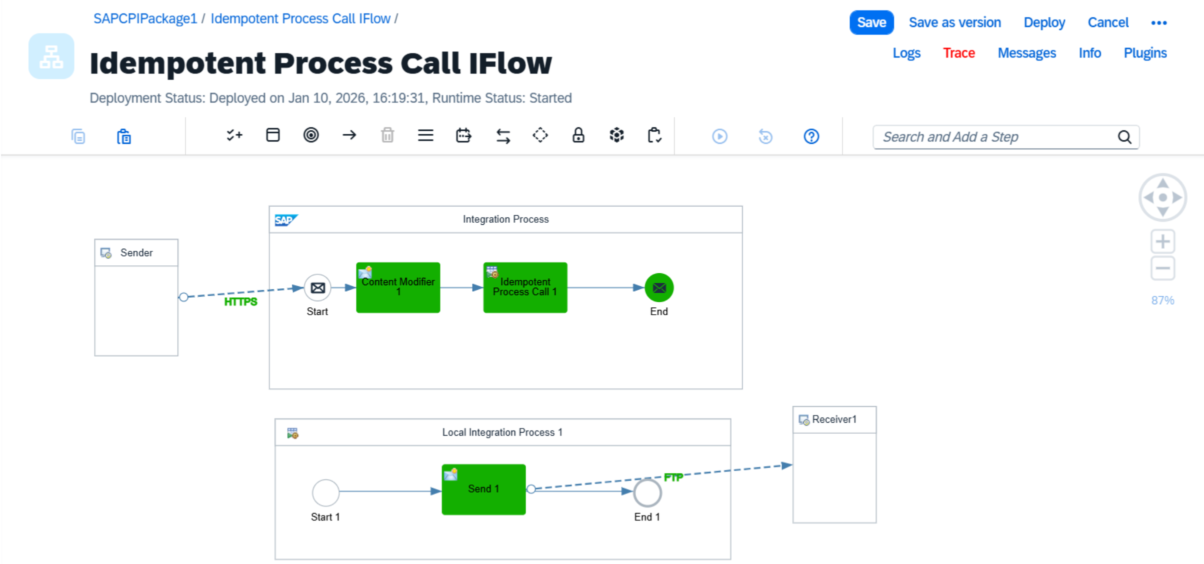
In cases where the **backend system is non-idempotent**, it cannot detect or manage duplicate messages on its own. Or if the HTTP method used is non-idempotent.
That's where the **middleware** must take responsibility.

- a **unique ID** is required to identify duplicates. In our scenario, the unique reference is the **Sales Order ID** included in the order XML.
- This ID serves as the key for the middleware to check whether a message has already been processed. If the middleware encounters this Sales Order ID for the first time, it allows the message to pass through to the backend system.
- If the same ID appears again, whether due to network retries, system errors, or users clicking "Submit" multiple times, the middleware can safely **ignore the duplicate message**.
- By implementing this check, the middleware ensures that the backend system receives each order **only once**, even if the message has been sent multiple times.

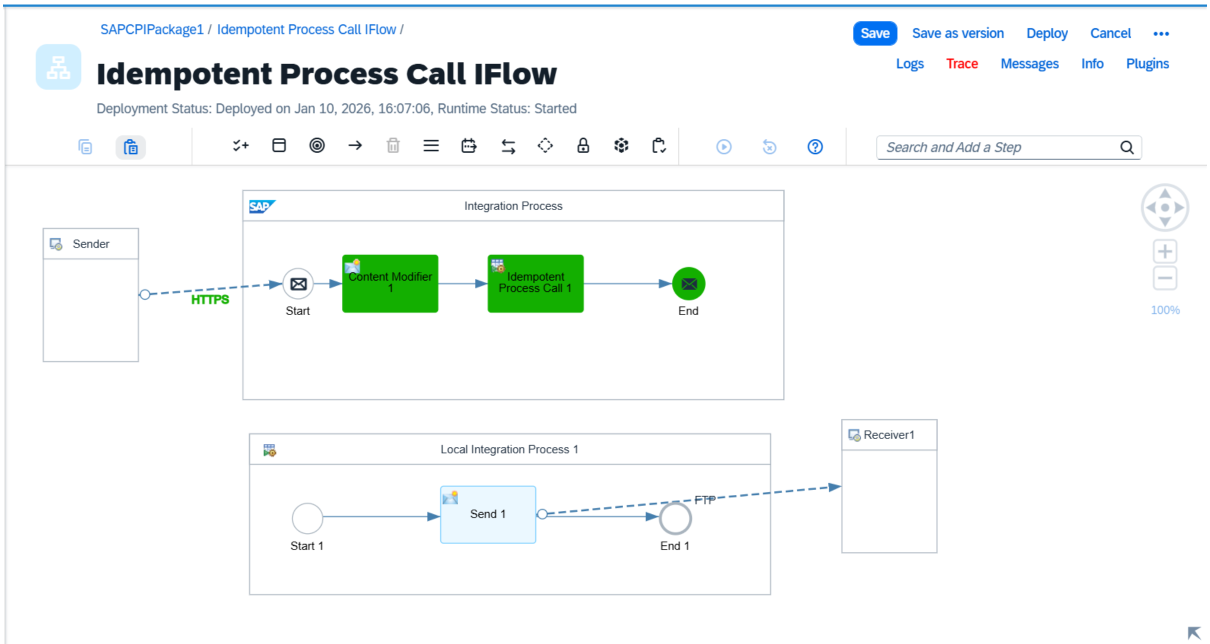
Prerequisites for Using the Idempotent Process Call in SAP Integration Suite

- One of the prerequisites for using the Idempotent Process Call in SAP Integration Suite is that we must have a unique ID to identify each message (or iteration).
- This ID is then used to detect duplicates and ignore any subsequent duplicate messages associated with that same ID.

2.Architechture Diagram



Message Processed successfully



Local process call skipped

3.Step-by-Step Implementation

- Incoming sales order messages are received and processed in the main integration flow.
- Extracted the Order ID from the payload using an XPath expression to uniquely identify each message.

- Configured an Idempotent Process Call using the extracted Message ID to ensure duplicate message handling.
- Enabled “Skip Process Call for Duplicates” option so that duplicate messages are automatically ignored by the system.
- Implemented a Local Integration Process responsible for forwarding valid and unique messages to the FTP receiver system.

Successful Test case: Message Processed successfully

- Tested the interface by sending a unique sales order, which was successfully processed and received by the FTP target system.
- Verified message processing using SAP Integration Suite Message Monitoring and Traces.

Failed Test case: Local Process call skipped

- Sent a duplicate sales order message and confirmed that the FTP receiver did not receive the message.
- Observed in the Message Processing Log that the Local Integration Process Call was skipped for the duplicate message.

4.Adapter Configuration:

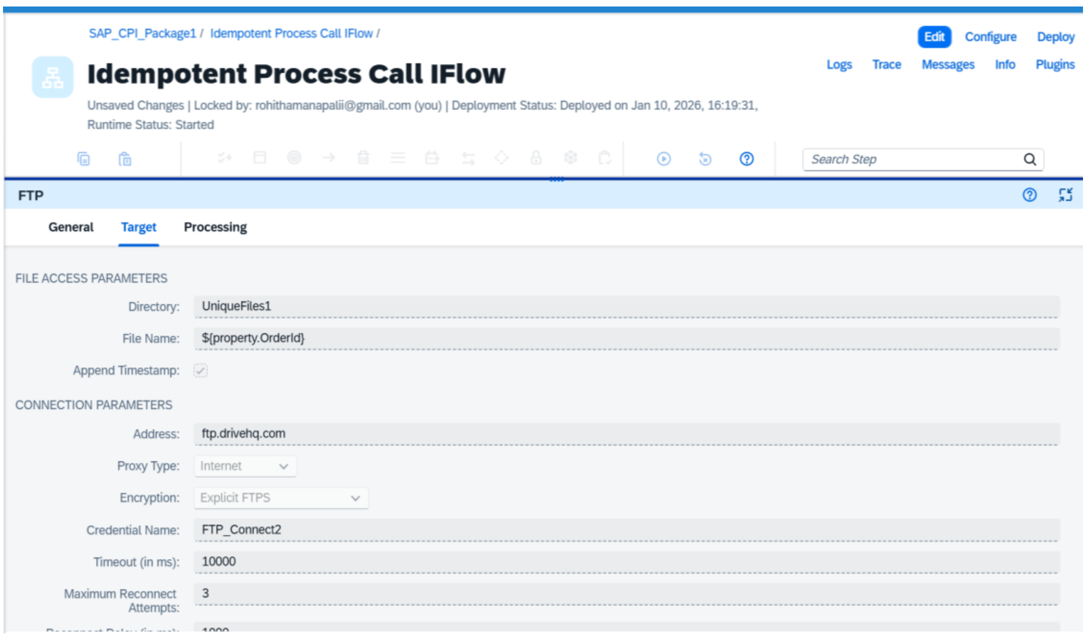
HTTPS Sender Adapter

Endpoint: https://b84a8864trial.it-cpitrial05-rt.cfapps.us10-001.hana.ondemand.com/http/idempotentCall

Method: POST

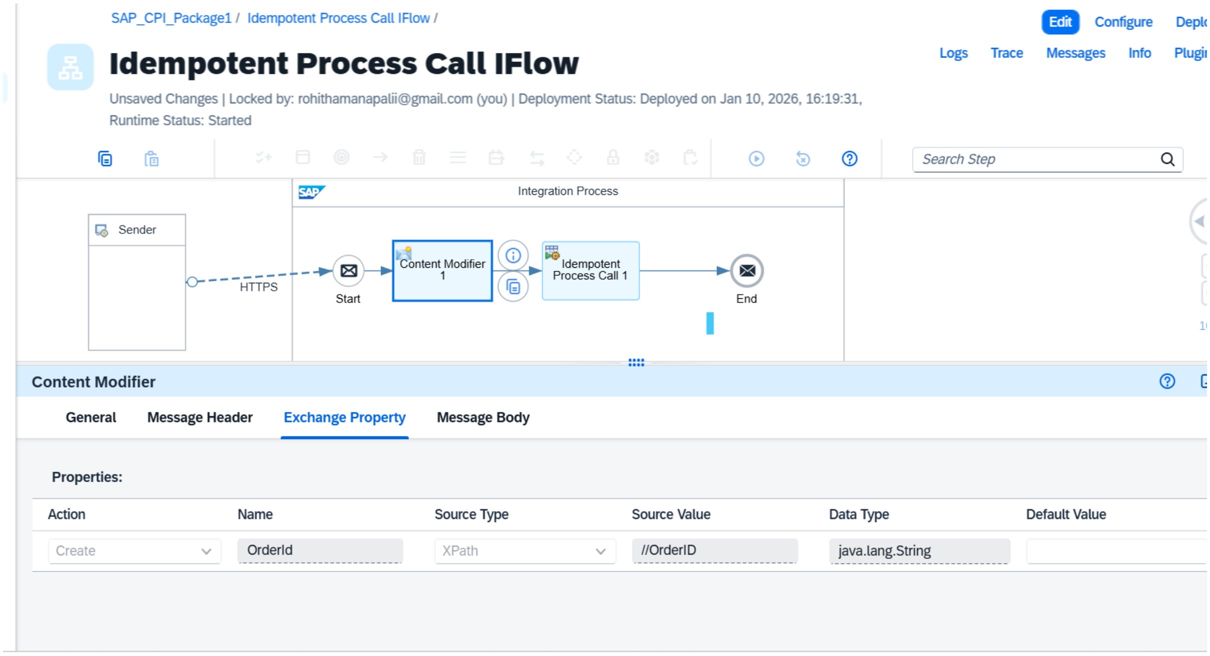
Authentication: Basic

FTP Receiver Adapter



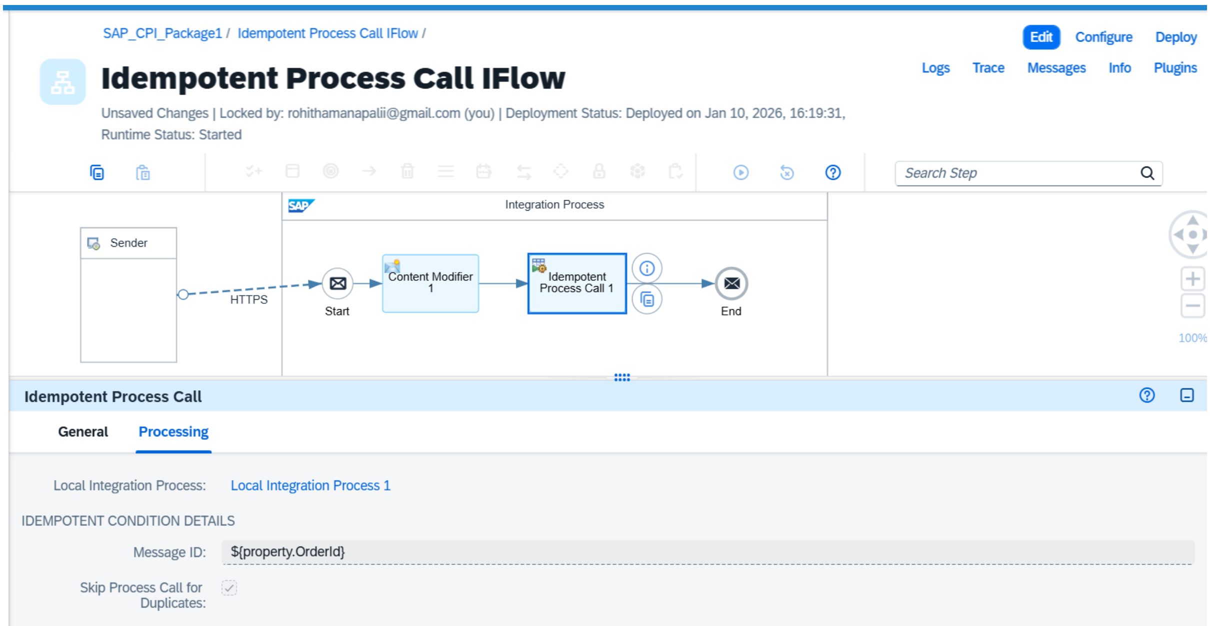
5.Integration Components:

Content Modifier:



Extract Unique ID from incoming payload

Idempotent Process Call:



6. Input XML

<?xml version="1.0" encoding="UTF-8" ?>

<Order>

```xml
<OrderID>10250</OrderID>
<CustomerID>VINET</CustomerID>
<EmployeeID>5</EmployeeID>
<OrderDate>1996-07-04T00:00:00Z</OrderDate>
<RequiredDate>1996-08-01T00:00:00Z</RequiredDate>
<ShippedDate>1996-07-16T00:00:00Z</ShippedDate>
<ShipVia>3</ShipVia>
<Freight>32.38</Freight>
<ShipName>Vins et alcools Chevalier</ShipName>
<ShipAddress>59 rue de l'Abbaye</ShipAddress>
<ShipCity>Reims</ShipCity>
<ShipRegion></ShipRegion>
<ShipPostalCode>51100</ShipPostalCode> <ShipCountry>France</ShipCountry>
</Order>
```


Idempotent Repository and Validity Period

- However, note that the idempotent repository data is retained for **90 days only**. After 90 days, the stored Message IDs are removed, meaning duplicate checks can only be performed within a 90-day window from the first message execution. In most real-world scenarios, this time frame is more than sufficient